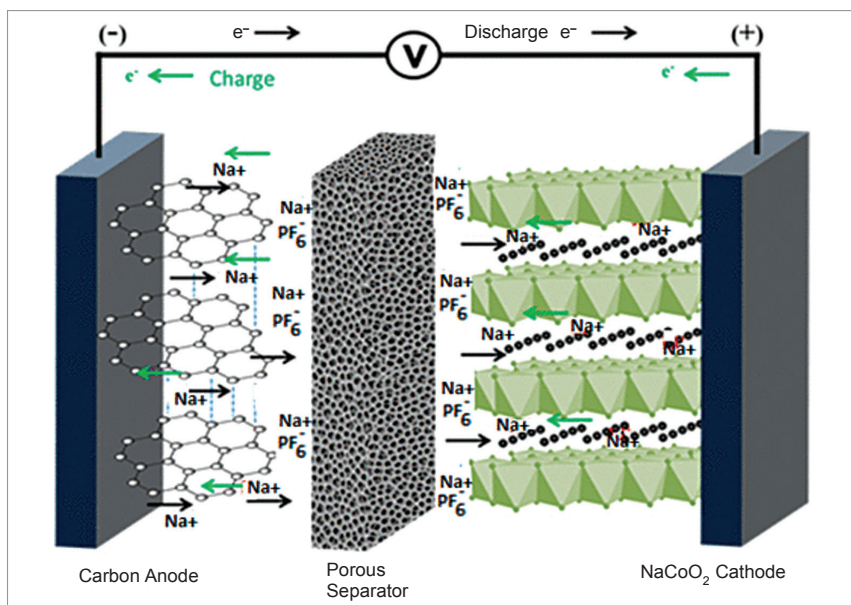




Schematic representation of a Na-ion battery cell (Credit: <https://pubs.acs.org>)

ing an average voltage of 3.7V during discharge. The development gained momentum only during the last decade as SIBs were found to be the most promising alternative to LIBs. Efforts are now underway to establish scalable production technology capable of meeting the commercial targets of a lifetime cost less than £0.1/kWh, deep discharge cycle life greater than 10,000 cycles, capital expenditure less than £250 (usable kWh installed), volumetric energy density greater

than 20kWh per cubic metre, and lifetime over 10 years. While initially, Japanese scientists made crucial advances, Chinese companies are now leading in commercialising the technology. It is expected that in the next two years, China will have nearly 95 percent of the world's capacity.

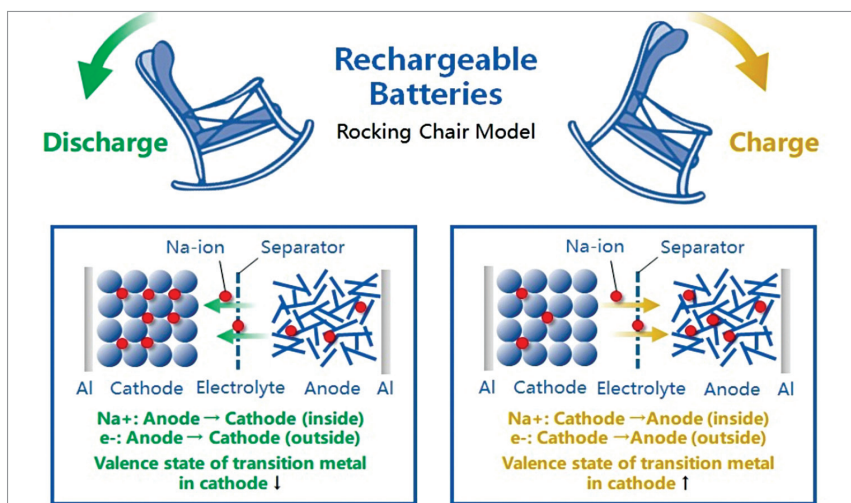
Recently, China-based HiNa Battery Technology demonstrated an electric test car using a 140Wh/kg SIB. The world's biggest battery manufacturer, CATL, has announced

that it will begin mass production in 2023. However, commercially available EVs using sodium-ion batteries are not yet available.

How SIBs work?

A sodium-ion battery is made up of an anode (not necessarily a sodium-based material), a cathode made of a sodium-containing material, a porous separator, liquid electrolyte containing dissociated sodium salts in polar protic or aprotic solvents, and two current collectors—one positive and one negative. Polar protic solvents, such as alcohols and water, have acidic protons, while polar aprotic solvents, like acetone and crown ethers, do not have acidic protons. The anode and cathode store the sodium, while the electrolyte carries charged ions from the anode to the cathode and vice versa through the separator. The movement of sodium ions creates free electrons in the anode, creating a charge at the positive current collector. The current then flows from the current collector through the device being powered by the battery, such as a smartphone, to the negative current collector. The separator blocks the flow of electrons inside the battery. During discharging, the anode releases sodium ions to the cathode, generating a flow of electrons from one side to the other. During charging, sodium ions move from the cathode to the anode, while electrons travel through the external circuit.

Electrode materials. Electrode materials for SIBs can be classified into three basic categories based on the reaction mechanism: insertion, alloying, and conversion. Insertion materials insert the host sodium ion (Na^+) into the interstitial space; alloying materials form an alloy with the host material during the electrochemical reaction; conversion materials react with host ions to form another compound that can be recovered upon reverse reaction.



SIBs can be explained using the rocking chair model, which describes the discharge and charge processes over a wide temperature range (Credit: <https://www.hinabattery.com>)